# A Single-Model Study of Change Detection Under Resolution Degradation: Simulated Degradation and Cross-Dataset Sentinel-2 Evaluation

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Abstract—Deep change detectors are often trained on high-resolution aerial imagery such as LEVIR-CD (0.5 m), while Sentinel-2 provides 10 m and 20 m bands. This paper reports a single-architecture case study of that gap. One MobileNetV2-based Siamese U-Net is trained once on LEVIR-CD and frozen. At native resolution it reaches pooled F1 87.06%, IoU 77.08%, precision 88.82% and recall 85.36% on 2,048 test pairs. We blur and downsample the same pairs to 1, 2, 4, 8 and 10 m, upsample them to $256 \times 256$, and re-evaluate without retraining. F1 is 70.81% at 1 m and 4.41% at 2 m; at 8–10 m it is 2.61%, recall is 1.32%, precision is about 99.5%, and only 0.068% of pixels are predicted changed. Among 935 changed test images, 924 (98.8%) receive no predicted change at 10 m. The simulation isolates blur and resampling and does not reproduce Sentinel-2 spectral, radiometric or atmospheric differences. We also evaluate the frozen CNN zero-shot on the ten official OSCD test cities: pooled F1 is 0.0007 with joint-percentile preprocessing and 0.0000 with reflectance. Train-selected spectral rules reach pooled F1 0.0782 (hardcoded NDVI/NDBI) and 0.1176 (percentile); Otsu predicts 67.35% of pixels changed and is summarized by overall accuracy 0.3439. These results do not establish spectral indices as a validated alternative. The test cities had been scored more than once before the reported run, so the evaluation is not an independent test of unseen cities. Only one architecture and one training run are tested.

Index Terms—change detection, remote sensing, resolution mismatch, domain shift, convolutional neural networks, Sentinel-2, spectral indices, NDVI, NDBI, OSCD

## I. Introduction

Automated change detection supports urban planning, disaster response and land-cover monitoring. Much recent progress uses deep networks trained and evaluated on high-resolution benchmarks such as LEVIR-CD, a dataset of bitemporal Google Earth imagery with change annotations [1]. Results on a benchmark's native imagery do not by themselves establish transfer to a different sensor, resolution or label definition.

Sentinel-2 is a multispectral imaging mission with bands sampled at 10 m, 20 m and 60 m spatial resolutions [2], [9]. A 10 m pixel is twenty times coarser linearly than a 0.5 m aerial pixel. The difference matters for small objects, mixed pixels and any decision that assumes native spatial detail.

This paper reports an empirical case study, not a new detector. We take one trained CNN, freeze it, and evaluate it in two ways. We apply controlled blur and resampling to the LEVIR-CD test pairs to vary simulated resolution while holding scene content fixed. We then evaluate the same frozen checkpoint zero-shot on the official OSCD Sentinel-2 urban-change split and compare it with train-selected spectral rules. The spectral rules are included as inspectable reference points, not as a validated replacement for learned detection. We report city-bootstrap intervals, trivial baselines and the prior use of the test split so that the strength of each statement is visible.

The contributions are:

- a controlled degradation study of one frozen CNN over 2,048 LEVIR-CD test pairs at six simulated resolutions, with pooled confusion counts and a per-image characterization at 10 m showing near-empty predictions rather than a rise in false positives;
- a zero-shot OSCD evaluation of the same CNN and train-selected NDVI/NDBI rules, with city-bootstrap intervals and trivial baselines; and
- an explicit disclosure that the OSCD test cities had been scored before the reported run, so the result is descriptive rather than an independent confirmation.

We do not claim a new architecture, a benchmark-leading score or a validated operational system.

## II. Related Work

CNN change detection. Fully convolutional Siamese networks provide an early end-to-end architecture for bitemporal change detection [3]. STANet introduced spatial-temporal attention and the LEVIR-CD dataset [1]; BIT later used transformer-based temporal feature fusion [4]. These benchmark results use different protocols from the cross-resolution evaluation reported here and are not treated as direct comparisons.

Transfer and resolution shift. Domain shift between training and deployment imagery can involve changes in sensor, geography, season, atmosphere and resolution. This paper isolates a simulated blur-and-resampling factor in one experiment and reports a separate real-data evaluation; it does not identify which factor causes the cross-dataset result.

Spectral indices and thresholding. NDVI is a vegetation index derived from red and near-infrared reflectance [5]. NDBI uses shortwave-infrared and near-infrared reflectance to highlight built-up areas [6]. At 10 m, mixed pixels can dilute both indices, so an index difference threshold is not resolution-independent. Otsu's method selects a threshold from a gray-level histogram without labels [7].

Benchmarks and evaluation hygiene. OSCD provides Sentinel-2 urban-change pairs and annotations for cross-city evaluation [8]. We report the prior scoring history of its test cities and do not present the reported run as a fresh, independent test. The contribution is a bounded empirical case study, not a priority claim.

## III. Software Context and Scope

The current repository contains a React/TypeScript browser application served by an Express service. It compares provider RGB imagery and uploaded RGB pairs. For uploaded pairs, users can select a normalized region of interest, inspect the images with synchronized zoom and pan, and view a thresholded RGB pixel-difference screen. Provider imagery may be a composite or mosaic rather than an acquisition matching the requested date. If a complete provider pair is unavailable, the application labels generated imagery as a demonstration. The current application does not run the CNN or multispectral NDVI/NDBI pipeline described in the supplied draft; its RGB results are screening signals, not validated change classifications.

The experiments in Sections IV–V are separate historical offline evaluations of a frozen CNN and spectral rules. They are not measurements of the current browser application. The saved per-image degradation counts, per-city OSCD confusion counts, and scripts for deriving summary metrics are included in the repository history and have been restored. Their aggregate metrics and paper tables were recomputed from those records for this revision. The original imagery, model checkpoint and complete run environment are not available here, so inference and the original evaluations could not be rerun. No claim about application accuracy or operational performance is made. The distinction between the historical experiments and the current software is essential to interpreting this work.

## IV. Controlled Resolution Degradation

### A. Protocol

The checkpoint is a MobileNetV2-based Siamese U-Net with a 3D-convolution fusion module. It was trained once on 256×256 LEVIR-CD patches with a combined Dice and binary cross-entropy loss, AdamW (learning rate $5\times10^{-4}$, weight decay $10^{-4}$), batch size 4 and cosine annealing. Training was configured for up to 100 epochs with early stopping (patience 10) and stopped after 27 epochs; the best validation IoU was 0.7205. The checkpoint was frozen for all experiments below.

We used the 2,048 LEVIR-CD test pairs. For degradation factor $f > 1$, each image was blurred with a PIL Gaussian filter of radius $f/2$ pixels, downsampled to $(256/f) \times (256/f)$ with LANCZOS resampling, and upsampled back to $256 \times 256$ with LANCZOS. At 10 m this is a $12 \times 12$ intermediate for a $256 \times 256$ input. Images were normalized with ImageNet statistics and passed to the frozen CNN with a 0.5 probability threshold. TP, FP and FN were counted per image, and Table I pools the counts over all 2,048 pairs at each level. The procedure holds scene content fixed and varies blur and sampling; it does not reproduce Sentinel-2 spectral bands, radiometric calibration, atmospheric correction, mixed-pixel composition or sensor noise.

TABLE I
CNN scores versus simulated resolution on the 2,048 LEVIR-CD test pairs. Values are computed from pooled TP/FP/FN counts, not averaged per image.

| Factor | Resolution (m) | Precision | Recall | F1 | IoU |
|---|---|---|---|---|---|
| 1 | 0.5 | 88.82 | 85.36 | 87.06 | 77.08 |
| 2 | 1 | 93.43 | 57.01 | 70.81 | 54.81 |
| 4 | 2 | 90.78 | 2.26 | 4.41 | 2.25 |
| 8 | 4 | 97.69 | 1.44 | 2.84 | 1.44 |
| 16 | 8 | 99.59 | 1.32 | 2.61 | 1.32 |
| 20 | 10 | 99.47 | 1.32 | 2.61 | 1.32 |

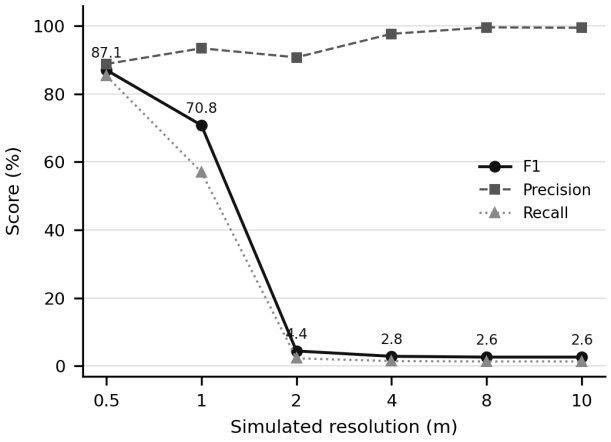



Fig. 1. F1, precision and recall versus simulated resolution on the 2,048 LEVIR-CD test pairs. F1 and recall collapse beyond 1 m; precision remains high at coarse levels, where very few pixels are predicted changed.

### B. Results

Table I and Fig. 1 report the pooled results. F1 is 87.06% at 0.5 m, falls to 70.81% at 1 m (a 16.25-point absolute

drop), and collapses to 4.41% at 2 m. It is 2.84% at 4 m and 2.61% at 8–10 m. Recall falls from 85.36% at 0.5 m to 57.01% at 1 m and 2.26% at 2 m, then 1.32% at 8–10 m. Precision is 88.82%, 93.43%, 90.78%, 97.69%, 99.59% and 99.47% across the six levels. The decline in F1 is therefore recall-dominated.

The high precision at coarse levels should not be read as useful detection. At 10 m only 91,006 of 134,217,728 pixels are predicted changed (0.068%), against 6,837,404 changed ground-truth pixels. Precision is computed from 90,527 true positives and 479 false positives, so it describes how little the model predicts rather than how well it localizes change.

### C. Prediction Behavior at 10 m

Of the 935 test images that contain ground-truth change, 924 (98.82%) receive no predicted change pixels at 10 m. The remaining 11 divide into 4 with IoU below 0.1, 5 between 0.1 and 0.5, and 2 at or above 0.5. These counts are descriptive, not mechanistic; they do not separate resolution loss from the other differences in the simulation.

## V. Real Sentinel-2 Evaluation on OSCD

### A. Data and Protocol

OSCD provides fourteen training cities and ten official test cities with pixel-level urban-change labels [8]. We use five bands aligned to the label grid: B02, B03, B04, B08 and B11. The pooled test set contains 3,077,936 pixels, of which 5.17% are labelled change. The benchmark layout used here does not include the Sentinel-2 Scene Classification Layer, so cloud and shadow exclusion is not evaluated.

Three spectral configurations were evaluated. The hardcoded rule flags a pixel when NDVI drops by at least 0.30 or NDBI rises by at least 0.30. The percentile rule thresholds the positive-change evidence at the 99.0th percentile. Otsu thresholds the same evidence without labels [7]. The hardcoded and percentile settings were selected on the fourteen training cities over fixed grids according to the saved code and project records; the selected hardcoded values lie at the edges of the search grid, while 99.0 is interior to the percentile grid. The saved v2 path contains no test-label fitting step, but the complete threshold-selection and run logs are unavailable, so this separation cannot be independently audited.

The CNN is evaluated zero-shot with two preprocessing variants. Its input is an RGB stack (B04, B03, B02). Reflectance scaling maps raw values to 0–255 by dividing by 10,000; joint-percentile scaling uses the 2nd and 98th percentiles of both dates combined. Scenes are processed in $256 \times 256$ tiles with 32-pixel overlap and averaged probabilities. Joint-percentile was selected on training cities; reflectance is the alternative.

Scoring history. The ten official test cities had been scored before the reported v2 run. An earlier preliminary run with a smaller threshold grid produced pooled F1 values of 0.167 (percentile), 0.104 (hardcoded), 0.095 (Otsu) and 0.0 (CNN). The total number of scoring events is not recorded. The v2 run re-evaluated those cities after the configuration was fixed on training cities. It is therefore not an independent test of unseen cities, and we do not present it as one.

### B. Results

Table II reports pooled results. The percentile rule has the highest pooled F1 among the spectral configurations (0.1176, 95% city-bootstrap CI 0.0540–0.1802), with precision 0.2419 and recall 0.0777. The hardcoded rule reaches F1 0.0782 (CI 0.0168–0.1846) and predicts 2.24% of pixels as changed against a true 5.17%. The paired difference in F1 between percentile and hardcoded is +0.0394, with a bootstrap interval of −0.0176 to +0.0778; the interval includes zero, so the ranking is not resolved at the 5% level. Both spectral rules under-predict change in aggregate. A random predictor at 5.17% prevalence has F1 0.0517; only the percentile interval lies entirely above it, narrowly. Otsu predicts 67.35% of pixels changed and obtains overall accuracy 0.3439, with precision 0.0513 and recall 0.6684. Its F1 is 0.0953 (CI 0.0495–0.1245), but its predicted fraction is so far above the true prevalence that F1 is not a useful standalone summary; overall accuracy is reported as its headline score.

The CNN variants are close to empty. The joint-percentile variant predicts 0.01% of pixels and reaches F1 0.0007 (CI 0.0000–0.0017); the reflectance variant predicts 0.00% and reaches F1 0.0000. Their overall accuracy is 0.9483, equal to the all-no-change accuracy on this test set. This is expected for a near-empty predictor on a 5.17% positive class and is not detection performance.

Training-city robustness. We also checked threshold stability without touching the test split. For each of the 14 training cities, the hardcoded and percentile configurations were selected on the other 13 training cities and applied to the held-out training city. Pooled held-out F1 was 0.0976 for the hardcoded rule (IoU 0.0513, precision 0.0838, recall 0.1168) and 0.1469 for the percentile rule (IoU 0.0793, precision 0.1544, recall 0.1401). The hardcoded selection sat on a search-grid edge in all 14 folds; the percentile selection did not. The hardcoded scores are therefore limited by the grid. This is a training-split robustness check, not a test-set score, and it adds no test-scoring event.

### C. City-Level Variation

Fig. 2 shows that the pooled numbers hide large city differences. Percentile F1 ranges from 0.0005 (saclay_w) to 0.2481 (chongqing); hardcoded F1 ranges from 0.0032 (dubai) to 0.3102 (chongqing). The joint-percentile CNN produces no predicted change in eight of ten cities and reaches at most 0.0024 (lasvegas); reflectance produces none. These ranges are descriptive. The per-city intervals in Fig. 2(a) resample pixels independently, ignoring spatial

TABLE II

Pooled results on the ten official OSCD test cities (3,077,936 pixels; true change 5.17%). F1 intervals are 95% percentile intervals from a city-level bootstrap (seed 0, 10,000 resamples); they reflect city sampling only. OA is shown for every method; Otsu's predicted fraction is 67.35%, so its OA is its headline score. Precision is reported as 0 by convention for methods with no positive predictions.

| Method | Pred. (%) | Precision | Recall | F1 | F1 95% CI | IoU | OA |
|---|---|---|---|---|---|---|---|
| Hardcoded NDVI/NDBI | 2.24 | 0.1292 | 0.0561 | 0.0782 | [0.0168, 0.1846] | 0.0407 | 0.9317 |
| Percentile 99.0 | 1.66 | 0.2419 | 0.0777 | 0.1176 | [0.0540, 0.1802] | 0.0625 | 0.9397 |
| Otsu | 67.35 | 0.0513 | 0.6684 | 0.0953 | [0.0495, 0.1245] | 0.0500 | 0.3439 |
| CNN joint percentile | 0.01 | 0.2983 | 0.0003 | 0.0007 | [0.0000, 0.0017] | 0.0003 | 0.9483 |
| CNN reflectance | 0.00 | 0.0000 | 0.0000 | 0.0000 | [0.0000, 0.0000] | 0.0000 | 0.9483 |
| Random at prevalence | 5.17 | 0.0517 | 0.0517 | 0.0517 | – | 0.0265 | – |
| All no change | 0.00 | – | 0.0000 | 0.0000 | – | 0.0000 | 0.9483 |

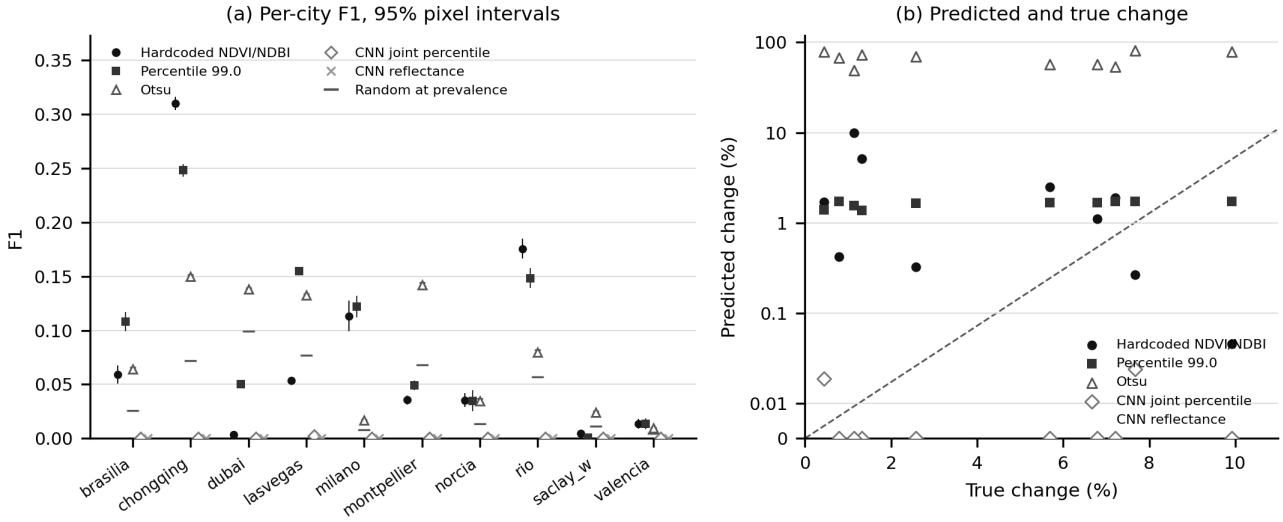



Fig. 2. OSCD v2 test results. (a) Per-city F1 with 95% pixel-bootstrap intervals; the short horizontal bars mark the random-at-prevalence expectation. F1 is shown for Otsu for completeness, although its headline score is overall accuracy. (b) Predicted versus true change fraction; the vertical axis is symmetric-logarithmic and the dashed line is predicted = true. The spectral rules sit mostly below the line, Otsu sits far above it, and the CNN variants sit near zero.

correlation; the pooled intervals resample the ten cities. Neither includes model-training or threshold-selection uncertainty.

The simulation isolates blur and resampling; the OSCD evaluation uses real data but confounds resolution with spectral, radiometric, atmospheric and geographic differences. No cloud or shadow mask is available in this OSCD layout, so cloud effects are also uncontrolled.

## VI. Discussion

The sharp transition between 1 m and 2 m is consistent with the loss of spatial information on which this architecture depends, but the experiment does not identify which features cause it. The practical implication is narrower and better supported: a model validated only at its native resolution can fail on coarser imagery, and a native-resolution benchmark score does not reveal that risk.

The coarse-resolution precision values illustrate a reporting problem. A near-empty predictor can obtain high precision and high overall accuracy while missing almost all change. Reporting F1 alone can also mislead when the predicted fraction is extreme, as with Otsu. Precision, recall, predicted fraction and true prevalence need to be read together, and overall accuracy is uninformative when the positive class is small.

For decision support, outputs should be treated as candidate evidence requiring human verification, not autonomous change polygons. The current browser application exposes RGB difference screening and source limitations; it does not produce validated land-cover classes or physical-area estimates. The spectral rules are transparent but weak: the percentile rule improves pooled F1 over the hardcoded rule, yet both under-predict change and vary widely by city. Their results do not establish NDVI/NDBI differencing as a validated low-resolution fallback. Otsu is not a drop-in fix; its high recall is bought with very low precision on this class balance.

## VII. Limitations

The degradation experiment applies blur and resampling to LEVIR-CD rather than acquiring real imagery at each resolution. It omits Sentinel-2 band differences, radiometric calibration, atmospheric correction, mixed

pixels and sensor noise. OSCD supplies real 10 m data, but the layout used here has ten test cities, no SCL layer and urban labels only. Only one architecture and one training run were tested; there was no retraining, fine-tuning, seed variation or architecture comparison, and the training seed was not recorded. The hardcoded thresholds sat on a search-grid edge in all 14 training folds (Sec. V-B), so the hardcoded scores are limited by the grid as well as by the data. No decision-threshold tuning was tested. The official test cities had been scored before the reported run, and the total number of scoring events is not recorded. The saved per-image degradation CSV and per-city OSCD confusion-count JSON are present, and their reported aggregate metrics can be reproduced from those records. The original imagery, model checkpoint and complete run environment are unavailable, so original inference cannot be independently rerun. No confidence intervals are reported for overall accuracy, and the per-city intervals use an independent-pixel approximation. The reported benchmark evaluates a historical offline pipeline and frozen checkpoint, not the current application's imagery retrieval, RGB screening, exports or user interface.

All test-set scores and per-image counts in this paper come from two saved files: degradation_results.csv and benchmarks/results/oscd_results_v2.json. The training-only robustness values come from benchmarks/results/oscd_loco_thresholds.json, produced without reading a test label. Their SHA-256 hashes and derivation instructions are recorded in the accompanying artifact guide. The figure/table generator reads the saved result files only and records their hashes in a manifest; it does not run model inference or rescore any test city for this paper.

## VIII. Conclusion

We reported a single-architecture case study in which a CNN with 87.06% F1 at native 0.5 m resolution falls to 70.81% at 1 m and 4.41% at 2 m under blur-and-resampling, and to 2.61% at 8–10 m with near-empty predictions. This specified blur-and-resampling simulation alone was sufficient to collapse this detector; it does not reproduce all differences of real Sentinel-2 data. A zero-shot OSCD evaluation of the same frozen CNN yields pooled F1 0.0007 or lower. Train-selected NDVI/NDBI rules also perform poorly (pooled F1 0.0782–0.1176), so the results do not establish spectral indices as a validated alternative. The test cities had been scored before the reported run. Resolution mismatch should be evaluated explicitly, and claims about low-resolution fallbacks require real-data validation on independent cities.

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